Kubernetes Troubleshooting Guide

This document lists common Kubernetes cluster problems, the likely causes behind them, the first thing to check, the sequence of places to inspect, and the usual fixes. Kubernetes troubleshooting is usually most effective when it starts at the failing resource, then moves outward to nodes, networking, storage, and finally control-plane components.[cite:3][cite:31]

General Troubleshooting Sequence

For most Kubernetes incidents, the fastest path is to inspect the failing object first, then move layer by layer through the cluster.[cite:3][cite:31]

1. Check object status with `kubectl get pods -A`, `kubectl get nodes`, or the relevant resource command.[cite:3][cite:31]
2. Run `kubectl describe` on the failing pod, node, PVC, service, or ingress to inspect events and conditions.[cite:3][cite:35]
3. Check logs with `kubectl logs`, and use `kubectl logs --previous` for restarting containers.[cite:35]
4. If the pod is healthy but traffic fails, inspect Services, Endpoints, Ingress, DNS, and NetworkPolicy objects.[cite:31][cite:34]
5. If scheduling or stability is the issue, inspect node conditions, kubelet health, and resource pressure.[cite:3][cite:19]
6. If volumes are involved, inspect PVCs, PVs, StorageClasses, and mount events.[cite:31]
7. If the issue affects the whole cluster, inspect API server access, scheduler behavior, controller-manager, and etcd health.[cite:3][cite:41]

Pod Startup and Runtime Problems

Error	First thing to do	Sequence of places to check	Common fixes
CrashLoopBackOff	Run `kubectl logs <pod> --previous` and `kubectl describe pod <pod>`.[cite:35]	Container logs, pod events, command/args, env vars, ConfigMaps, Secrets, probes, resource limits.[cite:35]	Fix application crash, missing config, bad dependency, failing probe, or insufficient memory/CPU.[cite:35][cite:12]
ImagePullBackOff / ErrImagePull	Run `kubectl describe pod <pod>` and inspect the image reference.[cite:12][cite:35]	Image name and tag, registry reachability, imagePullSecrets, namespace secret, DNS/network path to registry.[cite:12]	Correct image name or tag, fix credentials, or restore registry connectivity.[cite:12]

Error	First thing to do	Sequence of places to check	Common fixes
CreateContainerConfigError	Run `kubectl describe pod <pod>`.[cite:12]	Referenced Secrets, ConfigMaps, env vars, volume mounts, container command, security context.[cite:12]	Fix invalid references or bad container configuration.[cite:12]
ContainerCannotRun	Inspect `kubectl describe pod <pod>` and container logs if available.[cite:12]	Entry point, command/args, file paths inside image, permissions, image contents.[cite:12]	Correct startup command, file path, or container image build.[cite:12]
OOMKilled	Inspect pod status and restart reason with `kubectl describe pod <pod>`.[cite:12][cite:35]	Pod events, memory limits/requests, app memory profile, node memory pressure.[cite:35]	Increase memory limits, set realistic requests, or fix memory leak in the app.[cite:12][cite:35]
PodInitializing	Run `kubectl describe pod <pod>`.[cite:12]	Init containers, image pull state, volume mounts, startup dependencies, DNS/network access.[cite:12]	Fix failing init container, blocked dependency, or mount issue.[cite:12]
Readiness or Liveness probe failure	Inspect probe failures in `kubectl describe pod <pod>`.[cite:14][cite:35]	Probe path, port, timeout, initial delay, startup duration, backend dependency readiness.[cite:14]	Correct probe values, increase startup delay, or fix the application endpoint.[cite:14]

Scheduling Problems

Error	First thing to do	Sequence of places to check	Common fixes
Pending	Run `kubectl describe pod <pod>` and read scheduling events.[cite:12][cite:31]	Pod events, resource requests, node capacity, taints, tolerations, affinity rules, PVC state.[cite:12][cite:31]	Lower resource requests, add cluster capacity, fix placement rules, or resolve PVC issues.[cite:12][cite:31]
FailedScheduling	Inspect scheduler event messages in `kubectl describe pod <pod>`.[cite:12]	Scheduler events, node labels, taints, tolerations, anti-affinity, allocatable resources.[cite:12]	Add tolerations, relax affinity, relabel nodes, or scale nodes.[cite:12]
Unschedulable due to taints or cordon	Run `kubectl get nodes` and inspect node state.[cite:3]	Node schedulable status, taints, resource pressure, autoscaler status.[cite:3]	Uncordon nodes, remove incorrect taints, or add capacity.[cite:3]

Node Problems

Error	First thing to do	Sequence of places to check	Common fixes
NodeNotReady	Run `kubectl describe node <node>`.[cite:3][cite:19]	Node conditions, kubelet status, container runtime, CNI, heartbeat, certificates, resource pressure.[cite:19]	Restart kubelet or runtime, repair networking, renew certificates, or free resources.[cite:19]

Error	First thing to do	Sequence of places to check	Common fixes
DiskPressure	Inspect node conditions with `kubectl describe node <node>`.[cite:3][cite:19]	Node disk usage, container image cache, `/var/log`, ephemeral storage, evicted pods.[cite:19]	Clear disk space, rotate logs, prune images, or increase disk capacity.[cite:19]
MemoryPressure	Inspect node conditions and workload memory usage.[cite:3][cite:19]	Node allocatable memory, pod memory usage, restart patterns, oversized workloads.[cite:19]	Reschedule heavy workloads, adjust requests/limits, or add more nodes.[cite:19]
PIDPressure	Inspect node conditions and process usage.[cite:19]	Process count, runaway daemons, host-level leaks, noisy pods.[cite:19]	Stop leaking processes or increase node capacity.[cite:19]
Kubelet issues	Inspect node events and kubelet health.[cite:3][cite:19]	Kubelet service state, logs, certificates, runtime socket, network reachability.[cite:19]	Restart kubelet, fix config, or restore runtime connectivity.[cite:19]

Networking and Service Problems

Error	First thing to do	Sequence of places to check	Common fixes
Service unreachable	Run `kubectl get svc` and `kubectl get endpoints`.[cite:31]	Service selector, targetPort, pod labels, backing pods, kube-proxy or CNI state.[cite:31]	Fix selector mismatch, wrong port mapping, or backend pod health.[cite:31]
DNS failure	Check CoreDNS pods in `kube-system`.[cite:31]	CoreDNS pod status, CoreDNS logs, DNS config, NetworkPolicy, upstream connectivity.[cite:31]	Restart or repair CoreDNS and fix blocked network paths.[cite:31]
Pod-to-pod communication failure	Test connectivity from inside a pod.[cite:31]	CNI plugin, routes, NetworkPolicy rules, target pod readiness, service endpoints.[cite:31]	Repair CNI, allow traffic in policies, or fix backend readiness.[cite:31]
Ingress failure	Inspect ingress object and ingress controller logs.[cite:34]	Host/path rules, TLS secret, service backend, controller pods, external LB config.[cite:34]	Fix routing rules, backend service mapping, TLS secret, or controller setup.[cite:34]
NetworkPolicy blocking traffic	Compare intended traffic flow with active policies.[cite:31]	Namespace selectors, pod selectors, ingress rules, egress rules, default deny policies.[cite:31]	Add allow rules or correct selectors.[cite:31]

Storage Problems

Error	First thing to do	Sequence of places to check	Common fixes
PVC Pending	Run `kubectl get pvc` and `kubectl describe pvc <pvc>`.[cite:31]	StorageClass, PV availability, dynamic provisioner events, access mode, requested size.[cite:31]	Fix StorageClass, create matching PV, or restore dynamic provisioning.[cite:31]
Volume mount failure	Run `kubectl describe pod <pod>` and inspect mount events.[cite:31]	PVC binding, PV details, access mode, node attachment, backend storage system.[cite:31]	Correct PVC/PV mismatch, backend permissions, or storage availability.[cite:31]

Error	First thing to do	Sequence of places to check	Common fixes
StatefulSet storage issue	Inspect StatefulSet, related pods, and PVCs.[cite:7][cite:31]	Per-pod PVCs, storage class, pod identity, access mode, backend health.[cite:7][cite:31]	Repair volume templates, storage backend, or StatefulSet rollout.[cite:7]

Control Plane Problems

Error	First thing to do	Sequence of places to check	Common fixes
API server unavailable	Test access with `kubectl cluster-info` or API connectivity checks.[cite:3][cite:41]	Client kubeconfig, network path, auth, API server health, certificates, control-plane logs.[cite:41]	Fix kubeconfig, restore network access, or recover API server health.[cite:41]
etcd issue	Check whether cluster state reads and writes are failing.[cite:3]	etcd health, disk latency, quorum status, certificates, control-plane logs.[cite:3]	Restore quorum, repair storage, or recover from backup if needed.[cite:3]
Scheduler delay	Inspect pending pods and scheduler symptoms.[cite:3]	Pending queue, scheduler logs, cluster capacity, filtering constraints.[cite:3]	Add capacity or fix scheduler or pod placement rules.[cite:3]
Controller-manager failure	Inspect reconciliation failures across deployments, nodes, or jobs.[cite:3]	Controller-manager health, logs, control-plane pressure, leader election issues.[cite:3]	Restart or repair controller-manager and underlying control-plane issue.[cite:3]

Security and Access Problems

Error	First thing to do	Sequence of places to check	Common fixes
RBAC denied / Forbidden	Read the exact `forbidden` error text carefully.[cite:34]	Service account or user identity, verb, API group, Role, ClusterRole, RoleBinding, ClusterRoleBinding.[cite:34]	Add or correct the required permission binding.[cite:34]
ServiceAccount token issue	Inspect the pod spec and service account reference.[cite:7]	Namespace, token projection, secret mount, API access pattern, automount settings.[cite:7]	Attach the correct service account or restore token projection.[cite:7]
Admission webhook rejection	Read the webhook denial message in full.[cite:7]	Policy rule, webhook service, cert validity, namespace exclusions, webhook availability.[cite:7]	Fix the violating manifest or restore webhook health.[cite:7]

Quick Decision Tree

A practical way to approach Kubernetes incidents is to start from the symptom and follow the likely dependency chain.[cite:3][cite:31]

- If the pod is restarting, check logs, then config, then probes, then resources.[cite:35]
- If the pod is pending, check scheduler events, then node capacity, then taints and affinity, then PVC state.[cite:12][cite:31]
- If the app is running but unreachable, check Service, Endpoints, DNS, Ingress, and NetworkPolicy.[cite:31][cite:34]

- If many workloads fail at once, check nodes first, then control-plane availability.[cite:3][cite:19]
- If stateful apps fail, inspect PVC, PV, StorageClass, and mount events before changing the application.[cite:31]

Practical Command Order

The following command order works well in real clusters for first-pass investigation.[cite:3][cite:31][cite:35]

1. `kubectl get pods -A`
2. `kubectl describe pod <pod>`

```
3. `kubectl logs <pod>` or `kubectl logs <pod> --previous`
```

4. `kubectl get svc,endpoints,ingress -A`
5. `kubectl get nodes`
6. `kubectl describe node <node>`
7. `kubectl get pvc,pv,storageclass`
8. `kubectl cluster-info`

This sequence is effective because Kubernetes problems usually appear first as a pod symptom, while the root cause often sits in configuration, scheduling, storage, networking, or node health beneath that pod.[cite:3][cite:31]